VPC- Network-ACL

What is NACLs in VPC:

NACL (Network Access Control List) is a **security layer in VPC** that controls traffic at the **subnet level**.
It works like a **firewall for subnet** in AWS VPC.

In AWS, NACL is used to allow or deny traffic **before it reaches EC2**.

When NACL used

- Block IP
- Allow only specific network
- Extra security
- Subnet protection

Difference between Security Group and NACL

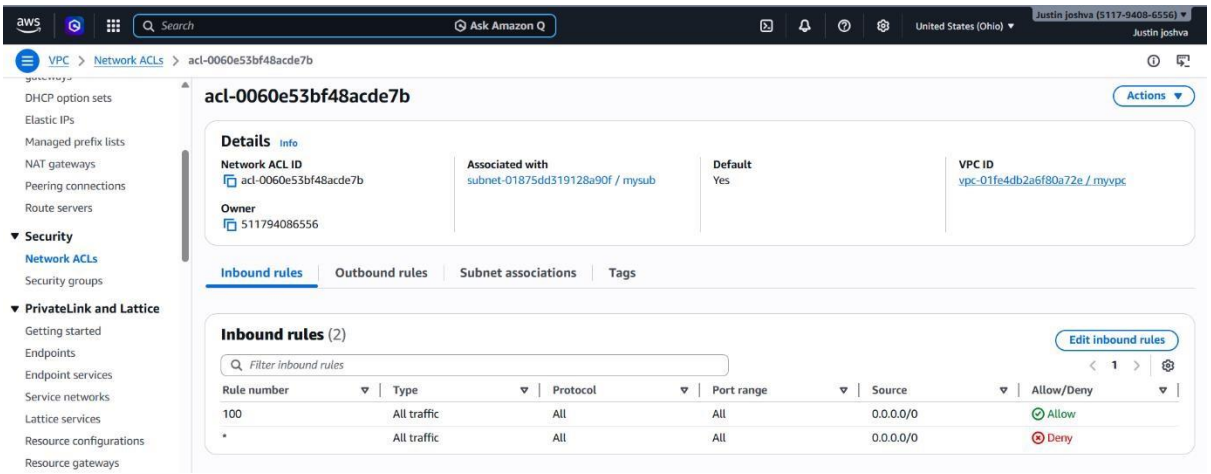
Feature	Security Group	NACL
Level	EC2	Subnet
Stateful	Yes	No
Allow	Yes	Yes
Deny	No	Yes
Rules	Simple	Number based

Stateless meaning

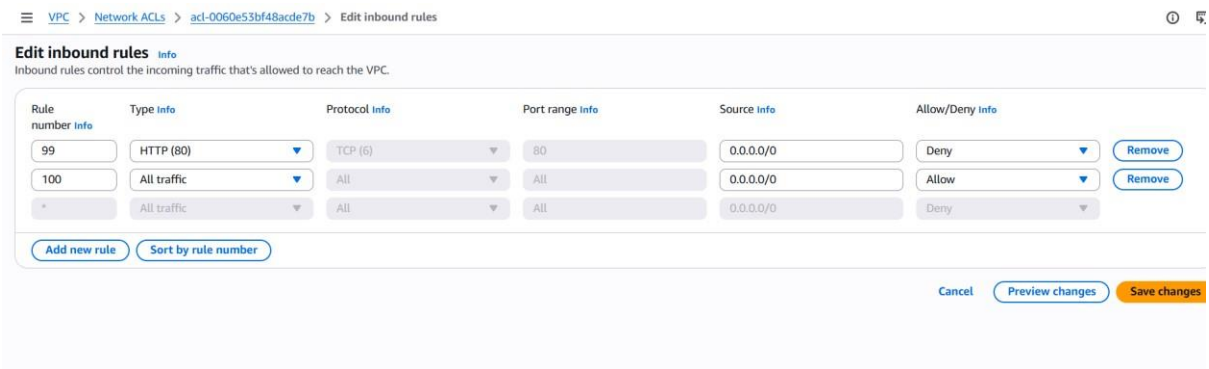
NACL is **stateless**, so you must allow both:

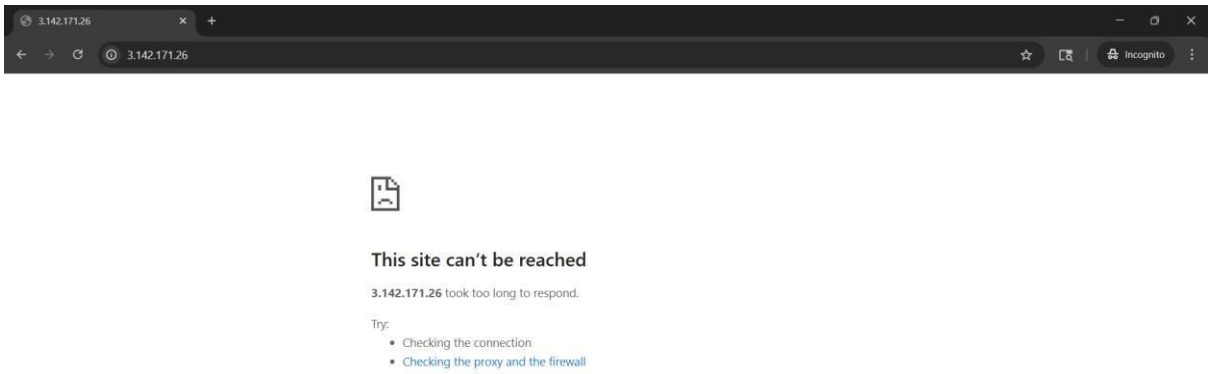
- Inbound
- Outbound

Here created the VPC, subnet, security group, internet gateway with route table and EC2 instance without edit the NALC and web-hosting .



Editing the NALC to deny the HTTP port



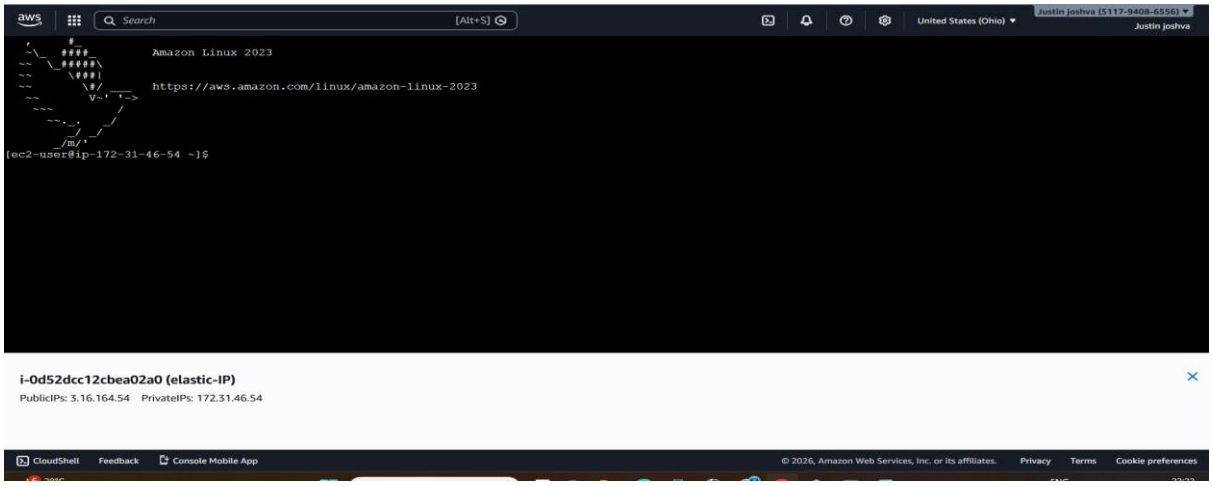


What is Elastic IP in AWS

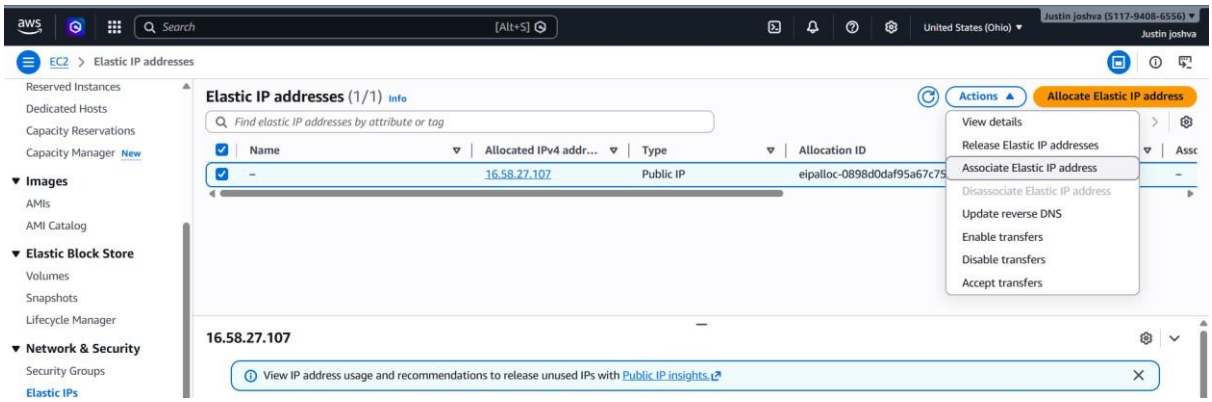
Elastic IP (EIP) is a static public IP address provided by AWS that you can assign to an EC2 instance.

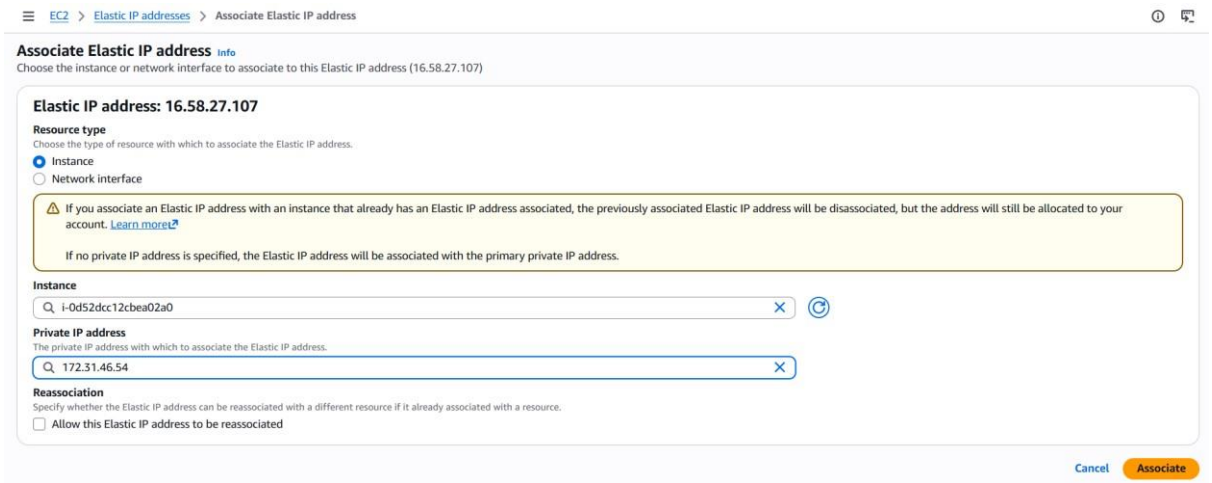
Normally, EC2 public IP changes when you stop and start the instance. Elastic IP stays the same.

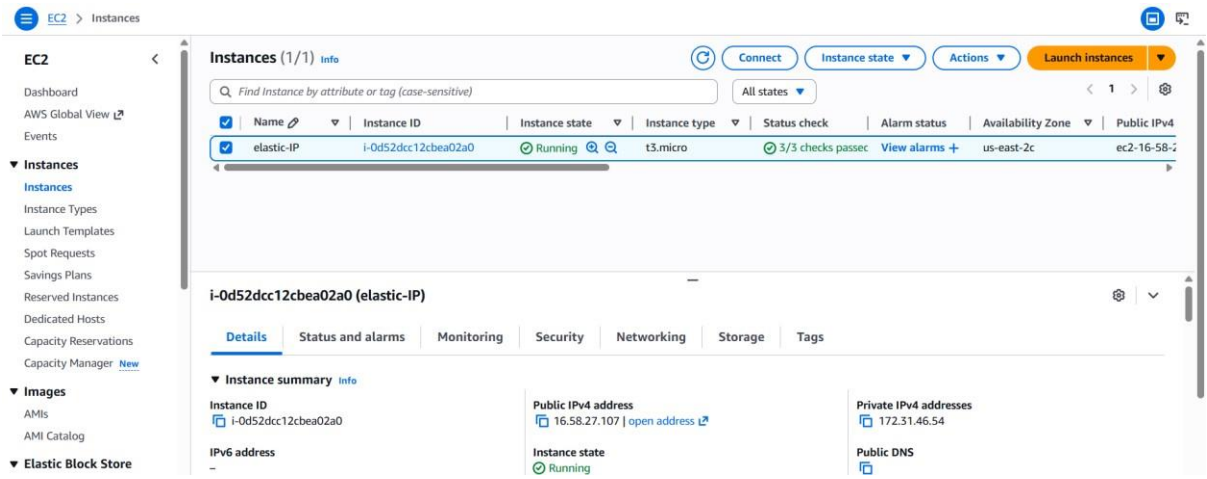
CREATED THE INSTANCE FOR THE ALLOCATE ELASTIC IP



ALLOCATE AND ASSOCIATE THE ELASTIC IP WITH EC2 INSTANCE


EC2 > Elastic IP addresses
Elastic IP addresses (1/1) Info
Actions
Allocate Elastic IP address
Find elastic IP addresses by attribute or tag
Name
Allocated IPv4 addr...
Type
Allocation ID
16.58.27.107
Public IP
View details
Release Elastic IP addresses
Associate Elastic IP address
Disassociate Elastic IP address
Update reverse DNS
Enable transfers
Disable transfers
Accept transfers
Reserved Instances
Dedicated Hosts
Capacity Reservations
Capacity Manager New
Images
AMIs
AMI Catalog
Elastic Block Store
Volumes
Snapshots
Lifecycle Manager
Network & Security
Security Groups
Elastic IPs
16.58.27.107
View IP address usage and recommendations to release unused IPs with Public IP insights.


EC2 > Elastic IP addresses > Associate Elastic IP address
Associate Elastic IP address Info
Choose the instance or network interface to associate to this Elastic IP address (16.58.27.107)
Elastic IP address: 16.58.27.107
Resource type
Choose the type of resource with which to associate the Elastic IP address.
Instance
Network interface
If you associate an Elastic IP address with an instance that already has an Elastic IP address associated, the previously associated Elastic IP address will be disassociated, but the address will still be allocated to your account. Learn more
If no private IP address is specified, the Elastic IP address will be associated with the primary private IP address.
Instance
i-0d52dcc12cbea02a0
Private IP address
The private IP address with which to associate the Elastic IP address.
172.31.46.54
Reassociation
Specify whether the Elastic IP address can be reassociated with a different resource if it already associated with a resource.
Allow this Elastic IP address to be reassociated
Cancel
Associate


EC2 > Instances
EC2
Dashboard
AWS Global View
Events
Instances
Instances
Instance Types
Launch Templates
Spot Requests
Savings Plans
Reserved Instances
Dedicated Hosts
Capacity Reservations
Capacity Manager New
Images
AMIs
AMI Catalog
Elastic Block Store
Instances (1/1) Info
Connect
Instance state
Actions
Launch instances
Find Instance by attribute or tag (case-sensitive)
All states
Name
Instance ID
Instance state
Instance type
Status check
Alarm status
Availability Zone
Public IPv4
elastic-IP
i-0d52dcc12cbea02a0
Running
t3.micro
3/3 checks passed
View alarms +
us-east-2c
i-0d52dcc12cbea02a0 (elastic-IP)
Details
Status and alarms
Monitoring
Security
Networking
Storage
Tags
Instance summary Info
Instance ID
i-0d52dcc12cbea02a0
Public IPv4 address
16.58.27.107 | open address
Private IPv4 addresses
172.31.46.54
IPv6 address
Instance state
Running
Public DNS